BEATRICE BAAH

Software & AI Engineer · Data & Security Background
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PROFILE

Software engineer who builds full-stack web applications and AI-powered products with React, TypeScript, Node.js, Python and Firebase, integrating Claude, Gemini and OpenAI APIs (including retrieval-augmented search). Came to software from five years of statistics and data-quality work in Ghana's local government and national statistical service, plus an MSc in Cybersecurity and Digital Forensics, and builds with the same priorities: correct data, secure handling of personal records, and products that real users adopt.

TECHNICAL SKILLS

Languages & frameworks: TypeScript, JavaScript, React, Node.js/Express, Python, Flutter (intermediate), SQL

Cloud & data: Firebase (Auth, Firestore, Cloud Functions, Hosting), Google Cloud Run, MongoDB, Power BI, STATA, Excel

AI integration: Claude, Gemini and OpenAI APIs; embeddings and vector search (RAG); AI-assisted development with Claude Code

Security: Network traffic analysis, anomaly detection (Isolation Forest), SIEM (Splunk, Wazuh), threat-intelligence APIs, Linux and Bash automation

SELECTED PROJECTS

Bekwai Tax Revenue: Municipal Revenue Management System — React · Firebase · Cloud Functions

- Designed and built, alone, a revenue system for the Bekwai Municipal Assembly: revenue collection, taxpayer database, electronic payment, receipt verification, and a dashboard of collected and outstanding revenue.
- Repository kept private because it holds taxpayers' personal and financial records; walkthrough available on request.

mySOC: Security Operations Center Dashboard — Python · Streamlit · Scapy

- Live packet capture with rule-based and Isolation Forest anomaly detection, threat-intelligence lookups (AbuseIPDB, AlienVault, VirusTotal), incident tracking and automated reports. github.com/Beatbaah/mySOC

ProAssess: AI Hiring Assessment Platform — React 19 · TypeScript · Firebase · Gemini

- Recruiters invite candidates by email; candidates take four timed reasoning tests; Gemini writes a score summary on submission. Built the candidate pipeline, a side-by-side shortlisting view and GDPR-compliant deletion of candidate data. github.com/Beatbaah/ProAssess

Dream Trip Planner: AI Travel Agent — Flutter · Node.js · Claude · OpenAI · Cloud Run

- Users describe a trip by voice, text or photos; Claude interprets the request, OpenAI embeddings retrieve similar trips from a vector database, and Claude writes a personalised summary per match. github.com/Beatbaah/dream-trip-planner

CivicMind AI: Municipal Intelligence Platform — React 19 · TypeScript · MongoDB · Gemini

- Tracks municipal revenue and department performance; Gemini turns figures into readable insight, with a rules-based fallback when no API key is available. github.com/Beatbaah/civicmind-ai

Excellent Susu: Savings Group Management App — React · Firebase

- Mobile-first app for rotating savings groups with four role levels, contributions, payouts, reminders, receipts and an audit trail. github.com/Beatbaah/susu-management-app

Client websites (live) — Firebase · Firestore

- Corporate World: office-wear e-commerce shop with authentication, cart, wishlist and checkout; Firestore security rules make orders immutable once placed (corporate-world-gh.web.app).
- Haske Spa & Events: booking site for a spa and event business, with a booking flow that routes into WhatsApp (haskespaandevents.web.app).

PROFESSIONAL EXPERIENCE

Municipal Statistician; Acting Head, Statistics Department — Nov 2023 – Present

Local Government Service, Bekwai Municipal Assembly, Ghana

- Compile and validate administrative data from all Assembly departments each quarter and report to the Ghana Statistical Service.
- Built Power BI dashboards used by 50+ stakeholders, reducing decision-making delays by about 40%.
- Prepare revenue trend analyses and forecasts for the Municipal Coordinating Director, and figures for the development plan and budget; supervise and train department staff.

Constituency Collation Officer (2024 General Election) — Nov – Dec 2024

Electoral Commission of Ghana, Bekwai Constituency

- Collated and verified presidential and parliamentary results from all polling stations, cross-checking result sheets for consistency.

Zonal Data Quality Monitor Dec 2023 – Apr 2024
Ghana Statistical Service, Bekwai / Amansie West

- Supervised data quality for 188 field staff across 288 enumeration zones in the Integrated Business Establishment Survey, using STATA and Excel checks to catch and correct errors.

Assistant Statistician Jan 2022 – Aug 2023
Local Government Service, Bekwai Municipal Assembly

- Analysed municipal datasets and produced Excel and Power BI reports for planning and policy.

Data Quality Monitor (2021 Population and Housing Census) Jun – Sep 2021
Ghana Statistical Service, Ga North Municipal

- Used STATA daily to find and resolve data errors across 186 enumeration areas.

Data Analyst (National Service) Sep 2019 – Sep 2020
Social Security and National Insurance Trust (SSNIT), Kumasi

- Validated and cleaned contribution data and prepared contribution reports for client companies.

EDUCATION

MSc Cybersecurity and Digital Forensics 2024 – 2026
Kwame Nkrumah University of Science and Technology (KNUST), Kumasi

- Thesis: CNN-based detection of adversarial-patch (spot evasion) attacks on ID-style face photos (TensorFlow/Keras).

BSc Statistics, Second Class Honours (Upper Division) 2015 – 2019
Kwame Nkrumah University of Science and Technology (KNUST), Kumasi

CERTIFICATIONS

Certified Cybersecurity Educator Professional (CCEP), 2025 · Google Cybersecurity Certificate, 2024 · IBM Respond and Mitigate Security Threats, 2024 · Mastercard Cybersecurity Job Simulation (Forage), 2024 · SQL Fundamentals (Dataquest), 2023 · Python Programming Foundations (Udacity), 2022 · Power BI in Data Analytics (Forage), 2022